

Math Routines Field Guide

Short, predictable structures that get every student talking about mathematics

TCE 412M · Guest session with Todd Edwards

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WHAT IS A ROUTINE?

"A mathematical routine is a short activity with a specific structure that allows students to engage in sense making through discourse, reasoning, and justification."

Newell & Orton (2018), "[Classroom Routines: An Invitation for Discourse](#)," *Teaching Children Mathematics* 25(2), 94–102.

Because the structure stays the same from day to day, students stop worrying about *what to do* and put their energy into *the mathematics*. A new image or prompt, but the same moves.

WARM IT UP, DON'T GIVE IT AWAY

A routine often opens a lesson. The best ones warm up the **same mathematics** the main task explores and let you hear how students already think, **without** handing over the strategy the task is meant to reveal.

Test: *after this warm-up, could a student still show me something of their own on the task?*

THE FOUR ROUTINES IN THE READING

Number Talk Images

ntimages.weebly.com

"How many do you see? How do you see them?"

Builds: flexible structure: grouping, arrays, multiplication, area.

Your move: annotate the image as students explain, so each strategy becomes visible to everyone.

Which One Doesn't Belong?

[Talking Math with Kids](#)

"Which one doesn't belong? Why?"

Builds: precise vocabulary and justification. Sets exist for shapes, numbers, graphs, and equations.

Your move: make sure everyone knows *any* of the four can be the answer. The argument is the point.

Estimation 180

estimation180.com

"What's a number that's too low? Too high? Your estimate, and why?"

Builds: number sense, reasonableness, and ratio and proportion reasoning.

Your move: collect the too-low and too-high bounds first. Every student can get in, whatever they know.

Would You Rather?

wouldyourathermath.com

"Would you rather...? Justify your choice."

Builds: comparison: unit rates, percents, deals, and area, with a real decision to make.

Your move: have students commit to a side before calculating, then let the math confirm or change their minds.

MORE ROUTINES TO EXPLORE

Same But Different: two images. How are they alike? How are they different?

Slow Reveal Graphs: a real graph revealed a piece at a time. Notice, wonder, revise.

Visual Patterns: growing figures. What would step 43 look like?

Open Middle: one answer, many paths, and a "best" one to hunt for.

Fraction Talks: what fraction of the shape is shaded?

Daily Routines Library: routines sorted by concept.

Make your own with nVoke

A free tool for building your own visual routines and hearing how someone else sees the figures.

1 · Build

A prompt pack:
WODB, Same & Different, Sort, I Spy

2 · Hand off

Give the device to a partner, or project it for the room

3 · Analyze

Read their answers and decide what to offer next

ohiomathteacher.github.io/applet-library/nvoke

Nothing is uploaded; everything stays on your device.

MOVES THAT MAKE ANY ROUTINE WORK

Think time first. Silent thumbs-up on the chest when ready, so no one gets rushed.

Turn and talk. Everyone says their idea out loud before anyone shares with the class.

Record publicly. Write or draw each idea where all can see it.

Revoice and connect. "Where do you see her $4 + 6$ in the picture?"

Ask for another way. "Did anyone see it differently?"

Keep it short. Five to ten minutes. The routine is the warm-up, not the lesson.

Routines and sites curated by the #MTBoS / #iteachmath community of teachers: Pierre Tranchemontagne (Number Talk Images), Christopher Danielson (WODB), Andrew Stadel (Estimation 180), John Stevens (Would You Rather?).